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Shared control with a novel dynamic authority allocation strategy based on game theory and driving safety field

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A B S T R A C T

This paper proposes a novel dynamic control authority allocation strategy based on the elliptic driving safety field for the game-based shared control, which aims to deal with human-automation conflicts during obstacle avoidance. The noncooperative game theory is used for modeling the interaction between a human driver and the trajectory-following controller, where the Nash equilibrium solution is derived by distributed model predictive control (DMPC) approach. In addition, the elliptic driving safety field is introduced to evaluate the driving risk by considering the driver-vehicle-road interactions under different driving contexts. The authority allocation ratio, denoting the ratio between the lateral position error weight of the human driver's cost function and the trajectory-following controller's, is determined by the shared dynamic authority allocation strategy designed based on the evaluated driving risk. More importantly, two case studies are provided to verify the elliptic driving safety field and the proposed dynamic authority allocation strategy on a straight or curve road. The results show that the proposed strategy can help the vehicle obtain great performance for obstacles avoidance, and the control authority is dynamically shifted between a human driver and the trajectory-following controller according to the evaluated driving risk of current driving condition.

1. Introduction

Automated driving has received growing attention for improving traffic safety and relieving the human driver from driving workload in recent years. However, there are various issues to be addressed before the automated driving systems are employed into commercialization, such as technical limitations [1], and social dilemma [2]. In addition, there exists the loss of situation awareness of the human driver because the driver is out-of-the-loop during automated driving [3]. Therefore, in order to resolve above problems, shared control is an alternative solution as the control authority of the vehicle is continuously shared by a human driver and the automated systems, which incorporates the capabilities of both control participants.

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The haptic shared control and steer-by-wire systems provide possibilities for shared control where a human driver and automated systems could control the vehicle simultaneously, and they are also proved to improve the driving tasks performance effectively [4,5]. Previous studies of shared control focus on the design of the shared controller with multiple constraints, such as model predictive control (MPC) [6], Takagi–Sugeno fuzzy control [7], and robust control [8]. These researches pay more attention to the effectiveness and vehicle safety performance of the shared controller by considering the human driver in the control loop, and insufficient attention to the human-automation interaction. However, the guideline for shared control agrees that there are continuous interaction and dynamic authority allocation between the automated systems and a driver [4,9].

In terms of the human-automation interaction of shared control, game theory has been widely used as it is an effective method to address the conflicts between two or more players of game theory [10]. Na et al. apply the noncooperative game theory to model the human-automation interaction for collision avoidance maneuver, and authors also discuss the difference between the open-loop Nash and Stackelberg equilibrium of game theory, which both could be solved by linear quadratic (LQ) and distributed model predictive control (DMPC) method [11–13]. Different from Na et al., Flad et al. adopt the moveme driver model to predict the driver behavior and propose an open-loop Stackelberg game-based cooperative advanced driver assistance system (ADAS) [14]. In order to explore comprehensive mathematical interaction model, Ji et al. study the closed-loop feedback Nash and Stackelberg strategy of game theory, and discuss differences between the closed-loop and open-loop information patterns [15]. Besides, the results in [15] show that the information pattern has little influence on Nash strategy. Research in [16] proposes a stochastic game-based shared control system by considering the uncertainty of the human driver behavior. However, the control authorities of both the human driver and the automated systems in the above game-based shared control approaches are fixed. Mars et al. also argue that the control authority of the automation should depend on the traffic environment [17]. Ludwig et al. propose a time-variant differential game-based cooperative controller to deal with the loss of driver's situation awareness by shifting the control authority from ADAS systems to a human driver [18,19]. But the approach proposed by Ludwig et al. has limitations on the application of scenarios since the function of control authority allocation only relates to the simulation time.

Inspired by Mars et al. and Ludwig et al., this paper addresses the problem of the control authority allocation under different driving conditions. In our study, there is an assumption that the human driver is under the normal driving status. The control authority should be transferred from the automated systems to the human driver to deal with dangerous driving scenarios, and on the contrary, the control authority will be transited to the automated systems to decrease the human driver's control workload under safe driving conditions. Therefore, a novel dynamic control authority allocation strategy based on the elliptic driving safety field is designed for obstacle avoidance in this paper. The concept of driving safety filed is proposed by Wang et al. [20,21] to represent the driving risk influenced by the driver-vehicle-road interactions, which is constructed by using the artificial field theory [22,23]. In our study, the elliptic driving safety field is introduced by taking the differences of the influence between the tangent and normal direction of the road lane on the driving risk into consideration. The driving risk under different scenarios could be evaluated by the elliptic driving safety field and it is also utilized to design the dynamic control allocation strategy by using lookup table method. Assume that the accurate information of surrounding objects and road lanes are obtained by the advanced sensor fusion technology [24,25]. The collision-free trajectory is planned by the optimal trajectory planning method based on natural cubic spline in real time [26], and it is also inputted into the trajectory-following controller designed by MPC approach to realize obstacles avoidance safely [27,28]. By referring to Na et al. [11], the noncooperative game is used to model the interaction between the trajectory-following controller and the human driver, and the DMPC method is adopted to obtain the Nash equilibrium solution of game theory.

The main contributions of our study are listed as follows:

(1) A novel dynamic control authority allocation strategy for the noncooperative Nash game-based shared control is proposed based on the elliptic driving safety field. To the authors' knowledge, the driving safety filed is the first attempt to be applied to the game-based shared control. Importantly, the control authority allocation adapts to the traffic environment condition to handle the human-automation conflicts during obstacle avoidance.
(2) The driving safety field is improved by considering the influence of different directions on the driving risk, therefore, the elliptic driving safety field is introduced. Simultaneously, the coordinate-transformation matrix between the earth coordinate and the local coordinate of the road center line is considered in the elliptic driving safety field to adapt to different road shapes.

This paper is organized as follows: The human-automation interaction modeled by the noncooperative game theory is introduced in Section 2. Section 3 describes the elliptic driving safety field and the evaluation of driving risk, which are based on the artificial field theory. Next, Section 4 introduces the dynamic control authority allocation strategy based on the evaluated driving risk. The case studies are given in Section 5, and Section 6 provides a discussion and conclusion of this paper.

2. Noncooperative game-based shared control

The human-automation interaction is modeled by using the noncooperative game theory, and the shared steering control problem is solved by the noncooperative Nash strategy. The DMPC method is useful and efficient for solving the Nash equilibrium of the game theory [13]. In this section, the shared control problem and the Nash equilibrium solution of the game theory will be briefly introduced.

2.1. Illustration of the driver-controller shared control system

The interaction between a driver and the trajectory-following controller is modeled by the noncooperative game theory, and the framework of the driver-controller noncooperative game control is illustrated in Fig. 1. Inspired by Na et al. [11], the trajectory-following controller compensates the human driver's control signal to decrease possibilities of potential accidents caused by the human driver's wrong actions, and the human driver should estimate the control signal of the controller if he/she still wants to control the vehicle by himself/herself. In our study, the dynamic interaction model is built, where the controller will control the vehicle to decrease driver's workload under safe condition, however, the driver with the normal driving status still needs to control the vehicle under dangerous conditions in the shared control system. Generally, a driver and the trajectory-following controller compensates for each other's control input signal. As a result, the two participants' control strategies converge to the Nash equilibrium of the noncooperative game theory. As shown in Fig. 1, $\boldsymbol{R}_1(k)$ is the collision avoidance path for the trajectory-following controller, which could be obtained by the optimal trajectory planning method based on natural cubic spline in real time [26]. $\boldsymbol{R}_2(k)$ refers to the human driver's previewed target path. $\boldsymbol{Q}_{1m}(k)$ and $\boldsymbol{Q}_{2m}(k)$ are the dynamic weights for the trajectory-following controller and a driver, respectively, which are determined based on the evaluated driving risk, and it will be introduced in section IV. Besides, the vehicle dynamics is influenced by both the driver and the controller. The two-DOF (degree of freedom) single-track bicycle vehicle model [29] is used, and the discrete form of the driver-controller system dynamics could be expressed as follow

$$\begin{aligned} \boldsymbol{x}(k+1) &= \mathbf{A}\boldsymbol{x}(k) + \mathbf{B}_1\delta_1(k) + \mathbf{B}_2\delta_2(k) \\ \boldsymbol{z}(k) &= \mathbf{C}\boldsymbol{x}(k) \end{aligned} \tag{1}$$

where $\boldsymbol{x}(k)$ is the state vector including the sideslip angle $\beta(k)$, yaw rate $\omega(k)$, lateral position $y(k)$, and yaw angle $\varphi(k)$. $\mathbf{A}$ is the state matrix of $\boldsymbol{x}(k)$, $\mathbf{B}_1$ and $\mathbf{B}_2$ refer to the input matrices of the trajectory-following controller and the human driver [11], respectively. $\mathbf{C}$ is the system output matrix, and $\boldsymbol{z}(k)$ is the system output vector consisting of the lateral position $y(k)$ and yaw angle $\varphi(k)$.

2.2. Noncooperative Nash strategy

The DMPC method is introduced by Rawlings and Mayne to solve the Nash strategy of the noncooperative game theory in [30]. In order to deal with the driver-controller noncooperative game-based shared control problem, there are three parts for DMPC approach: prediction equations, cost functions, and Nash strategy, which will be introduced as follows.

2.2.1. Prediction equations

The two-DOF single-track bicycle vehicle model presented in Eq. (1) is employed into the DMPC framework to predict the future outputs based on the information at current time step. The predicted output of the driver-controller system at next time step $(k+1)$ could be evaluated by iterating

$$\boldsymbol{z}(k+1) = \mathbf{C}\boldsymbol{x}(k+1) = \mathbf{C}(\mathbf{A}\boldsymbol{x}(k) + \mathbf{B}_1\delta_1(k) + \mathbf{B}_2\delta_2(k)) \tag{2}$$

Therefore, the outputs of next N_p prediction horizon time step can be predicted by iterating the above equation N_p times. Besides, with the next N_u control horizon steps of control signals, generally, $N_u \leqslant N_p$, and the prediction equations could be denoted as

$$\boldsymbol{Z}(k) = \boldsymbol{\Phi}\boldsymbol{x}(k) + \boldsymbol{\Theta}_1\boldsymbol{\delta}_1(k) + \boldsymbol{\Theta}_2\boldsymbol{\delta}_2(k) \tag{3}$$

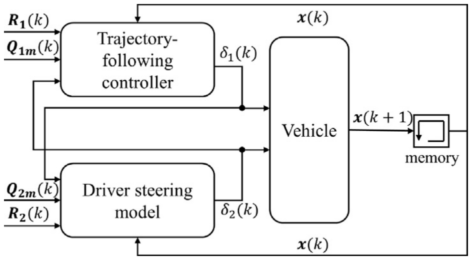

Fig. 1. Framework of the driver-controller noncooperative game-based shared control.

where

$$\boldsymbol{Z}(k) = \begin{bmatrix} \boldsymbol{z}(k+1) \\ \boldsymbol{z}(k+2) \\ \vdots \\ \boldsymbol{z}(k+N_p) \end{bmatrix}, \ \boldsymbol{\Phi} = \begin{bmatrix} \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{N_p} \end{bmatrix}$$

$$\boldsymbol{\delta}_1(k) = \begin{bmatrix} \delta_1(k) \\ \delta_1(k+1) \\ \vdots \\ \delta_1(k+N_u-1) \end{bmatrix}, \ \boldsymbol{\delta}_2(k) = \begin{bmatrix} \delta_2(k) \\ \delta_2(k+1) \\ \vdots \\ \delta_2(k+N_u-1) \end{bmatrix}$$

$$\boldsymbol{\Theta}_1 = \begin{bmatrix} \mathbf{CB}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{CAB}_1 & \mathbf{CB}_1 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{CA}^{N_p-1}\mathbf{B}_1 & \mathbf{CA}^{N_p-2}\mathbf{B}_1 & \cdots & \mathbf{CA}^{N_p-N_u}\mathbf{B}_1 \end{bmatrix}$$

and

$$\boldsymbol{\Theta}_2 = \begin{bmatrix} \mathbf{CB}_2 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{CAB}_2 & \mathbf{CB}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{CA}^{N_p-1}\mathbf{B}_2 & \mathbf{CA}^{N_p-2}\mathbf{B}_2 & \cdots & \mathbf{CA}^{N_p-N_u}\mathbf{B}_2 \end{bmatrix}$$

2.2.2. Cost functions

In the framework of the driver-controller noncooperative game-based shared control, the objective of each participant is to follow their own desired paths as closely as possible. Therefore, the path following error between the desired target path and the actual path should be penalized in the design of cost functions for the trajectory-following controller and the human driver. Besides, the control signal should be considered to minimize the workload. The cost functions of the trajectory-following controller and the human driver are respectively defined as

$$J_1(k) = \|\boldsymbol{Z}(k) - \boldsymbol{R}_1(k)\|^2_{\mathbf{Q}_{1m}(k)} + \|\boldsymbol{\delta}_1(k)\|^2 \tag{4}$$

and

$$J_2(k) = \|\boldsymbol{Z}(k) - \boldsymbol{R}_2(k)\|^2_{\mathbf{Q}_{2m}(k)} + \|\boldsymbol{\delta}_2(k)\|^2 \tag{5}$$

with

$\mathbf{Q}_{1m}(k) = diag\{\mathbf{Q}_1(k)\},\ \mathbf{Q}_{2m}(k) = diag\{\mathbf{Q}_2(k)\}$where $\mathbf{Q}_1(k)$ and $\mathbf{Q}_2(k)$ are the dynamic weight matrices for path following errors, which both include the weights of the lateral position error q_y and the yaw angle error q_φ, and they will be discussed in the dynamic control authority allocation strategy in section IV.

2.2.3. Nash strategy

The Nash equilibrium is defined as the solution that none of the participants can benefit more by unilaterally changing their control actions when it is applied into the driver-controller shared control system [30]. Therefore, the Nash equilibrium is derived by simultaneously solving the following coupled control problems

$$\min_{\delta_1^*} J_1(k) \text{ and } \min_{\delta_2^*} J_2(k) \tag{6a}$$

subject to

$$\boldsymbol{Z}(k) = \boldsymbol{\Phi}\boldsymbol{x}(k) + \boldsymbol{\Theta}_1\boldsymbol{\delta}_1(k) + \boldsymbol{\Theta}_2\boldsymbol{\delta}_2(k) \tag{6b}$$

where (δ_1^*, δ_2^*) is the Nash equilibrium solution for the driver-controller noncooperative game-based shared control problem. Besides, the zero-input tracking errors $\boldsymbol{\varepsilon}_i(k)$ of the trajectory-following controller and the human driver are defined as

$$\boldsymbol{\varepsilon}_i(k) = \boldsymbol{R}_i(k) - \boldsymbol{\Phi}\boldsymbol{x}(k) - \boldsymbol{\Theta}_j\boldsymbol{\delta}_j(k) \tag{7}$$

where $i+j=3$, and $i,j \in \{1,2\}$. By substituting Eq. (7) into Eqs. (4) and (5), the cost functions of the trajectory-following controller and the human driver could be expressed as

$$J_i(k) = \left\| \begin{bmatrix} \boldsymbol{S}_i\{\boldsymbol{\Theta}_i\boldsymbol{\delta}_i(k) - \boldsymbol{\varepsilon}_i(k)\} \\ \boldsymbol{\delta}_i(k) \end{bmatrix} \right\|^2 \tag{8}$$

where $\boldsymbol{S}_i^T\boldsymbol{S}_i = \boldsymbol{Q}_{im}(k)$. Above optimization problem with the form of the least-squares could be well solved by QR algorithm [31]. The optimal control signal of the trajectory-following controller and a driver are calculated by

$$\begin{bmatrix} \boldsymbol{\delta}_1^* \\ \boldsymbol{\delta}_2^* \end{bmatrix} = \begin{bmatrix} \boldsymbol{K}_1^f & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{K}_2^f \end{bmatrix} \begin{bmatrix} \boldsymbol{\varepsilon}_1(k) \\ \boldsymbol{\varepsilon}_2(k) \end{bmatrix} \tag{9}$$

with

$$\boldsymbol{K}_i^f(k) = [\boldsymbol{S}_i\boldsymbol{\Theta}_i\mathbf{I}] \setminus [\boldsymbol{S}_i\mathbf{0}]$$

where $\mathbf{I}$ is an identity matrix. By observing Eqs. (7) and (9), the control signal sequence of the human driver $\boldsymbol{\delta}_2(k)$ has influence on the optimal control signal sequence of the trajectory-following controller $\boldsymbol{\delta}_1^*(k)$ and vice versa. In order to solve above problem, the convex iteration method is adopted [30]. Therefore, the Nash solution of the driver-controller shared control system are determined by

$$\begin{bmatrix} \boldsymbol{\delta}_1^*(k) \\ \boldsymbol{\delta}_2^*(k) \end{bmatrix} = \boldsymbol{K}^{Nash}(k) \begin{bmatrix} \boldsymbol{x}_1(k) \\ \boldsymbol{x}_2(k) \end{bmatrix} \tag{10}$$

with

$$\boldsymbol{K}^{Nash}(k) = \begin{bmatrix} (\mathbf{I} - \boldsymbol{\Lambda}_1\boldsymbol{\Lambda}_2)^{-1}\boldsymbol{\Gamma}_1 & (\mathbf{I} - \boldsymbol{\Lambda}_1\boldsymbol{\Lambda}_2)^{-1}\boldsymbol{\Lambda}_1\boldsymbol{\Gamma}_2 \\ (\mathbf{I} - \boldsymbol{\Lambda}_2\boldsymbol{\Lambda}_1)^{-1}\boldsymbol{\Lambda}_2\boldsymbol{\Gamma}_1 & (\mathbf{I} - \boldsymbol{\Lambda}_2\boldsymbol{\Lambda}_1)^{-1}\boldsymbol{\Gamma}_2 \end{bmatrix}$$

where

$$\boldsymbol{x}_i(k) = [\boldsymbol{x}(k) \quad \boldsymbol{R}_i(k)]^T,$$

$$\boldsymbol{\Gamma}_i = \begin{bmatrix} -\boldsymbol{K}_i^f(k)\boldsymbol{\Phi} & \boldsymbol{K}_i^f(k) \end{bmatrix}, \ \boldsymbol{\Lambda}_i = \boldsymbol{K}_i^f(k)\boldsymbol{\Theta}_j$$

3. Elliptic driving safety field

Driving safety filed is practical to assess the driving risk of the host vehicle, which considers the influence of the driver-vehicle-road interactions on the driving safety [20,21]. In our study, the driving safety field is constructed with the ellipse shape, and it consists of the potential filed, the kinetic filed, and the behavior filed. In this section, the elliptic driving safety filed is briefly described.

3.1. Potential field

The potential field is an artificial filed denoting the influence of the non-moving objects on the driving safety. The non-moving objects could be divided into two categories.

(1) The non-moving objects with the potential collision possibility, such as static obstacle vehicles, roadblocks on the road.
(2) The non-moving objects imposing the behavior constraints for the human driver, such as traffic signs, road boundaries.

Therefore, the potential field consists of two parts in our study, namely, the obstacles potential and the road potential. The potential field will be described in detail as follows.

3.1.1. Obstacle potential

The purpose of the obstacle potential is to assess the collision risk between the host vehicle and around non-moving obstacle vehicles. The shorter relative distance between the host vehicle and the non-moving obstacle is, the greater collision possibility and the higher driving risk would be. Simultaneously, Takashi summarizes that the human driver fails to execute the lane-changing maneuver to avoid the obstacle ahead when time to collision (TTC) is lower than 6s and 2s is the margin of TTC [32]. It means that the human driver deserves more room in the tangent direction of the road lane to avoid obstacles than in the normal direction of the road lane since the lane width is smaller than the desired lane-changing distance. Therefore, we enlarge the influence of the tangent direction in our study. The elliptic obstacle potential of a non-moving obstacle at (x_o, y_o) is shown in Fig. 2, denoted as

$$\boldsymbol{E}_{R_o} = \frac{M_o R_o}{|\boldsymbol{r}_o^*|} \frac{\boldsymbol{r}_o}{|\boldsymbol{r}_o|} \tag{11}$$

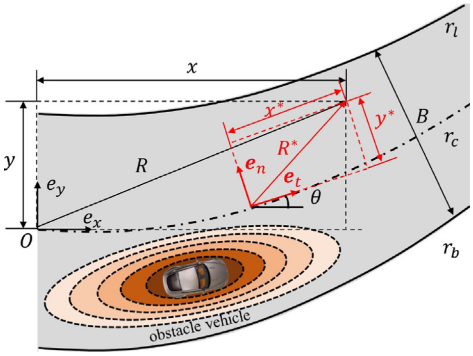

Fig. 2. Illustration of the elliptic obstacles potential and the road geometry. $\boldsymbol{e}_x$ and $\boldsymbol{e}_y$ are the unit vectors of the earth coordinate, $\boldsymbol{e}_n$ and $\boldsymbol{e}_t$ refer to the unit vectors of the local coordinate of the road centerline, $\boldsymbol{r}_l$ and $\boldsymbol{r}_b$ mean the road upper and bottom boundary, respectively, and $\boldsymbol{r}_c$ denotes the road center line.

with

$$\boldsymbol{r}_o^* = \begin{pmatrix} \frac{x^*-x_o^*}{a} & \frac{y^*-y_o^*}{b} \end{pmatrix} \tag{12}$$

where M_o is the virtual mass of the non-moving obstacle that is related to the vehicle mass m_o and its velocity [21], R_o refers to the road condition influence factor related to road conditions [20]. $\boldsymbol{r}_o = (x - x_o, y - y_o)$ means the distance vector, where (x, y) and (x_o, y_o) are the position vector of the host vehicle and the obstacle vehicle at the earth coordinate, respectively. $\boldsymbol{r}_o^*$ is the enlarged distance vector. $a$and $b$are the scaling factors of the tangent and normal direction, which could be tuned according to environment. In our study, $a$and $b$are determined by

$$a = |v_e - v_o|t_s, \;\; b = \tau \tag{13}$$

where v_e and v_o refer to the longitudinal velocity of the host vehicle and the obstacle vehicle, respectively. $t_s = 2s$ is the margin of TTC, and τ means the radius of safety circle [22]. (x^*, y^*) and (x_o^*, y_o^*) represent the position vector of the host vehicle and obstacle vehicle at the local coordinate of the road center line, as shown in Fig. 2. The coordinate-transformation matrix is given by

$$\begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \tag{14}$$

where θ is the angle between the earth coordinate and the local coordinate of the road center line. For more details of the road geometry, the readers could refer to [22]. Especially, $\boldsymbol{r}_o^*$ could be eliminated on the straight road, denoted as

$$\boldsymbol{r}_o^* = \begin{pmatrix} \frac{x-x_o}{a} & \frac{y-y_o}{b} \end{pmatrix} \tag{15}$$

3.1.2. Road potential

The second category of the potential filed, the road potential, doesn't lead directly to the collision, and only impose the behavior constraints on the driver. However, there are potential risks if the human driver violates the road constraints, such as lane departure. In our study, the road boundary constraints are considered, and its aim is to keep the host vehicle running on the road. The closer the vehicle is to the road boundaries, the higher the driving risk would be and the larger the road potential reaches. Simultaneously, the growing ratio of the road potential becomes greater with the relative distance decreasing. Therefore, the expression of the road potential at (x_a, y_a) is determined by

$$\boldsymbol{E}_{R_{a,q}} = \eta_q R_a \ln\left(\frac{B}{2} - |\boldsymbol{r}_a|\right) \frac{\boldsymbol{r}_a}{|\boldsymbol{r}_a|} \tag{16}$$

where η_q is the scaling factors of the road boundaries, and $q \in \{l, b\}$ represent the upper boundary and bottom boundary, respectively. $B$is the road width, and $\boldsymbol{r}_a = (x - x_{a,q}, y - y_{a,q})$ denotes the distance vector between the vehicle to the road boundaries. Especially, the curvy road is designed by the clothoid curve, and the expression of a curvy road center line could be denoted as

$$r_c(x) = y_0 + \delta_0 x + \frac{1}{2}c_0 x^2 + \frac{1}{6}c_1 x^3 \tag{17}$$

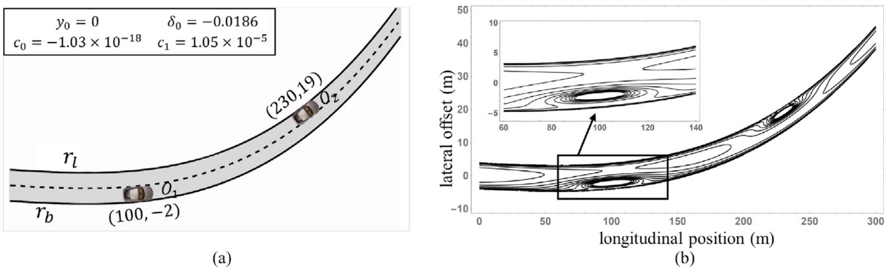

Fig. 3. Illustration of potential field: (a) Designed scenario including two non-moving obstacle vehicles and road boundaries, and (b) Corresponding potential filed.

Table 1
Parameters of The Elliptic Driving Safety Field.

Symbol	Description	Value(Units)
m_o	Mass of obstacle vehicles	1864(kg)
R_i	Road condition influence factor	1(–)
t_s	Margin of time to collision	2(s)
τ	Radius of safety circle	2(m)
η_q	Scaling factor of road border	−150(–)
B	Road width	8(m)
k_1	Scaling weight of kinetic field	0.005(–)
k_2	Scaling weight of field force	0.005(–)
DR_i	Driving risk factor of behavior field	0.5(–)

where y_0 means the initial lateral position. δ_0 is the initial heading direction of the curvy road. c_0 and c_1 refer to the initial curvature and its changing ratio, respectively.

Moreover, an example is provided to illustrate the potential field constructed by several non-moving objects (including non-moving vehicle obstacles and curvy road boundaries), which is shown in Fig. 3. The parameters of the potential field are provided in Table 1 where all the parameters of the elliptic driving safety field are listed. Assume that the host vehicle runs at speed of 20 m/s. As depicted in Fig. 3(a), the parameters of a curvy road are presented in the rectangle located at the top and left corner. Besides, there are two static obstacles, named as O_1 and O_2, and their lateral positions are at the middle on the left-hand lane and the right-hand lane, respectively. Their initial longitudinal positions are 100 m and 230 m far from the original point, respectively. From the Fig. 3(a), and the partial enlarged detail, the distribution of the driving risk and the elliptic potential fields of the non-moving obstacle vehicles could be easily observed. From the elliptic obstacles potential, the results show that the shapes of two static obstacle potentials can be adapted to the curvy road. The influence of the tangent direction is greatly larger than the normal direction under the condition that the value of $a$is larger than the value of b, and it means that there needs enough longitudinal distance for a driver to realize obstacle avoidance under a low risk level.

3.2. Kinetic field

The kinetic filed is an artificial filed denoting the influence of moving objects on the driving safety. For constructing the kinetic filed, there are several features of the moving objects considered.

(1) The shorter relative distance between the host vehicle and the moving obstacle is, the greater collision possibility and the higher driving risk would be. Simultaneously, the elliptic potential filed is adopted by considering the influence of the tangent and normal direction as described in previous part.
(2) The kinetic filed is also influenced by the characteristics of the relative movement between the host vehicle and the moving objects. At the same relative distance between the host vehicle and the moving object, the driving risk reaches the highest value when the host vehicle approaches from the front of the moving obstacle and reaches the lowest value when the host vehicle approaches from the rear of the moving obstacle.

Therefore, the elliptic kinetic filed of the moving object at (x_v, y_v) is determined by [20]

$$\boldsymbol{E}_v = \frac{M_v R_v}{|\boldsymbol{r}_v^*|} \frac{\boldsymbol{r}_v}{|\boldsymbol{r}_v|} \exp(k_1 v_v \cos\theta_v) \tag{18}$$

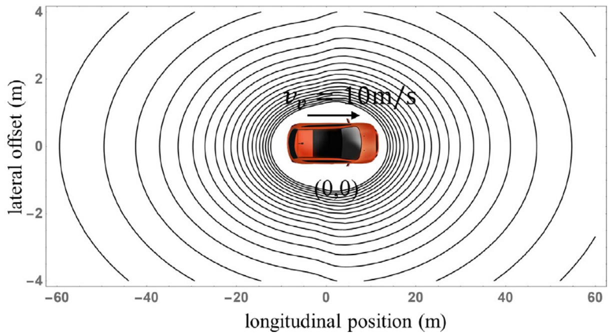

Fig. 4. Illustration of a moving obstacle vehicle and its corresponding kinetic filed.

where M_v is the virtual mass of the moving object, $\boldsymbol{r}_v = (x - x_v, y - y_v)$ refers to the distance vector between the host vehicle and the moving object, $\boldsymbol{r}_v^*$ is the enlarged distance vector, which is obtained by Eq. (12), k_1 is the scaling weight constant value greater than zero, v_v represents the velocity of the moving object, and θ_v is the angle between $\boldsymbol{r}_v$ and the velocity of the moving object.

The kinetic field constructed by Eq. (18) is illustrated in Fig. 4. The parameters of the kinetic field are listed in Table 1. Assume that the host vehicle moves at the constant velocity as 20 m/s. There is a moving obstacle vehicle runs at 10 m/s along the longitudinal direction, and its initial position at the original point. Different from the potential filed shown in Fig. 3, the kinetic filed increases under the influence of the direction of $\boldsymbol{r}_v$ and the moving object velocity in Fig. 4. If the direction of $\boldsymbol{r}_v$ coincides with the direction of the moving object velocity, the driving risk grows faster than the condition that the direction of $\boldsymbol{r}_v$ is contrary with the direction of the moving object velocity.

3.3. Behavior field

The behavior filed is an artificial filed denoting the influence of the characteristics of driver behaviors on the driving safety. For example, the driving risk provided by the behavior filed of an unskilled driver is always larger than that provided by the behavior field of a skilled driver, because the driving risk caused by the former driver is often higher than that caused by the latter driver. However, the driving risk caused by the behavior field to the surroundings is delivered by the moving vehicle driven by a driver. Therefore, the behavior field of a driver could be denoted as the product of the kinetic field of the host vehicle and a driving risk factor at (x_c, y_c)

$$\boldsymbol{E}_D = \boldsymbol{E}_v DR_c \tag{19}$$

where DR_c is the driving risk factor determined by the psychology and physiology, driving skill level, cognition, and traffic violations of a driver [20]. $\boldsymbol{E}_v$ is the kinetic field of the host vehicle, and the direction of $\boldsymbol{E}_D$ is same as $\boldsymbol{E}_v$. The larger value of $\boldsymbol{E}_D$ reflects the higher risk caused by the human driver's behavior.

3.4. Evaluation of driving risk

As mentioned previously, the driving safety field $\boldsymbol{E}_s$ is constructed by the potential field, the kinetic field, and the behavior field, denoted as

$$\boldsymbol{E}_s = \boldsymbol{E}_p + \boldsymbol{E}_v + \boldsymbol{E}_D \tag{20}$$

Besides, in order to denote the driving risk on the host vehicle, the field force is adopted, which is influenced by the field strength according to the above three fields, road condition, vehicle moving state, and behavior characteristics of a driver [20]. The field force $\boldsymbol{F}_{ri}$ on the host vehicle $i$at (x_i, y_i) is determined by

$$\boldsymbol{F}_{ri} = \boldsymbol{E}_{si} M_i [R_i \exp(-k_2 v_i \cos\theta_i)(1 + DR_i)] \tag{21}$$

where $\boldsymbol{E}_{si}$ represents the field strength vector of the driving safety field at $(x_i, y_i), M_i$ is the virtual mass of the host vehicle i, R_i is the road condition influencing factor, k_2 refers to the scaling weight constant value, v_i is the velocity of the host vehicle, θ_i means the angle between the angle of $\boldsymbol{E}_{si}$ and the direction of v_i, and DR_i represents the driving risk factor of the driver of the host vehicle i. The larger absolute value the field force $|\boldsymbol{F}_{ri}|$ is, the higher driving risk of the host vehicle would be.

4. Shared dynamic authority allocation strategy

The key part of the driver-controller shared control system is the dynamic authority allocation strategy between the trajectory-following controller and a human driver. The strategy is studied based on the evaluated driving risk, and it will be discussed detailedly in this section.

4.1. Dynamic authority allocation strategy

The objective of the dynamic authority allocation strategy is to adjust the control authority of the human driver or the controller according to the driving risk conditions. As presented in Eq. (6), the optimal shared control signal is calculated by solving the Nash equilibrium of the game theory. The interest of the control authority can be indicated by the penalization of the deviation to the reference path. The higher values of the path-error weight matrices $Q_1(k)$ or $Q_2(k)$ at each time step are, the larger effort the human driver or the controller would pay to follow their desired path. Therefore, the dynamic authority allocation could be achieved by adjusting the values of the path-error weight matrices of the human driver and the controller at their objective functions in real time.

In our study, the path-error weight matrices of the trajectory-following controller and a driver are defined in Eqs. (4) and (5), respectively, which include both the weights of the lateral position error q_y and the yaw angle error q_φ. However, Na et al. point out that there is little influence on the human driver's and the controller's steading-state steering control signals by changing the weight of yaw angle error q_φ[11]. The weight of the lateral position error will be discussed in our dynamic authority allocation strategy. Obviously, the higher ratio of the lateral position error weight between a driver and the trajectory-following controller is, the larger proportion of the control authority the human driver occupies and the closer the vehicle trajectory is to the target path of the human driver, and vice versa [11,12]. Importantly, the dynamic authority allocation strategy of the game theory is designed based on the driving risk in our study, and when the host vehicle runs under the higher driving risk conditions, the human driver should occupy more control authority in the shared control system. Therefore, the ratio between the human driver and the controller grows with the increasing of the evaluated driving risk, and the dynamic authority allocation strategy is defined as

$$\boldsymbol{Q}_2(k) = \lambda(|\boldsymbol{F}_r(k)|)\boldsymbol{Q}_1(k) \tag{22}$$

where λ is the authority allocation ratio related to the evaluated driving risk, and it will be introduced in the following part.

4.2. Authority allocation ratio

In our study, the authority allocation ratio is designed based on the driving risk, which is evaluated by the elliptic driving safety field. The basic design principles of the authority allocation ratio are summarized that the more dangerous the environment gets, the larger the authority allocation ratio should be and the higher control authority the human driver occupies. Therefore, the driving risk $|\boldsymbol{F}_r(k)|$ is selected as the input variable of the function of the authority allocation ratio. However, when the trajectory-following controller occupies the large proportion of the control authority, the authority allocation ratio λ is much less than 1 but greater than zero. On the contrary, λ is much greater than 1 when the human driver occupies larger control authority. By considering above characteristics of λ, the natural logarithm value of the authority allocation ratio $\ln(\lambda)$ is chosen as the output variable. The fuzzy control method is an alternative choice for the design of the authority allocation ratio as the driving risk level could be described by the fuzzy language [33]. However, using the lookup table method is more effective for a SI-SO (single input–single output) system.

To design a suitable strategy of the authority allocation ratio, the key step of using lookup table method is to determine nodes values of the input and output variables. Takashi summarizes that when TTC was 6s or less, the human driver would cancel the lane-changing maneuver to avoid obstacles, and 2s is the margin [32]. Besides, Wang et al. recommend that 4s of TTC is a warning time criterion [21]. Therefore, the driving risks at the positions of 6s, 4s, and 2s of TTC are calculated by Eq. (21), and they are used for the design of nodes values of the driving risk. Moreover, the condition that there are no obstacles on the road and the host vehicle keeps running on the right-hand lane is seen as the safe condition. And the driving risk of that condition calculated by Eq. (21) is also chosen for the design of driving risk's nodes value. The basic principle of the authority allocation ratio is that it increases with the driving risk. The nodes value of $\ln(\lambda)$ are designed by referring to [11,12] and simulation and experiment experience. Besides, the detailed information of the lookup table is listed in Table 2.

4.3. Selecting the weight setting of the controller

In the dynamic authority allocation strategy, the path-error weight of the human driver's and the controller's objective function should be determined at each time step. The authority allocation ratio λ can be obtained from the lookup table method. And the weight of the human driver is determined based on the weight of the controller. However, the weight of the controller is preset parameter in our strategy. Thus, the influence of the preset weight of the controller will be discussed and the best parameter will be chosen in this part.

Table 2
The Look-up Table of Authority Allocation Ratio.

$\lvert \boldsymbol{F}_r \rvert$	−400	-350	−300	−250	−200	-150	-100	−50	0	50
$\ln(\lambda)$	−4.27	−4.27	−4.27	−4.25	−3.56	−2.82	−2.35	−2.01	−1.78	−1.42
$\lvert \boldsymbol{F}_r \rvert$	100	150	200	250	300	350	400	450	500	550
$\ln(\lambda)$	−0.83	−0.42	0.00	0.21	0.63	1.17	1.61	1.83	2.21	2.34
$\lvert \boldsymbol{F}_r \rvert$	600	650	700	750	800	850	900	950	1000	
$\ln(\lambda)$	2.56	2.82	3.34	3.83	4.23	4.32	4.67	4.67	4.67	

4.3.1. Performance index

In order to evaluate the performance of different weights of the controller, the performance index is designed in our study, and defined as

$$PI = \varepsilon_1 \overline{PI_{DR}} + \varepsilon_2 \overline{PI_{CW}} + \varepsilon_3 \overline{PI_{MC}} \tag{23}$$

with

$$\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 1 \tag{24}$$

where $\overline{PI_{DR}}, \overline{PI_{CW}}$, and $\overline{PI_{MC}}$ are the normalized values of evaluated performance of the driving risk (DR), control workload (CW), and the maximum curvature (MC) of the trajectory, respectively.$\varepsilon_1, \varepsilon_2$, and ε_3 are the weights of three evaluated terms, respectively. Specially, $\varepsilon_1 = \frac{1}{2}, \varepsilon_2 = \frac{1}{4}$, and $\varepsilon_3 = \frac{1}{4}$ are selected since the more attention is paid on the driving safety in our study when compared with other two items. The min–max normalization method is used for normalizing the performance index items among the tested groups. The first evaluated term $\overline{PI_{DR}}$ reflects the sum of driving risk of the host vehicle under current weight, and denoted as

$$\overline{PI_{DR}} = \int_0^T \lvert \boldsymbol{F}_r \rvert dt \tag{25}$$

where $T$is the simulation time. The evaluated term $\overline{PI_{CW}}$ is to assess the control workload of the system, including the sum of the control signal and its changing rate, and it is given by

$$\overline{PI_{CW}} = \int_o^T \left(\delta + \dot{\delta} \right) dt \tag{26}$$

In order to evaluate the comfort of the lateral movement, smoothness criterion of the vehicle trajectory should be considered in the last term $\overline{PI_{MC}}$, which is the maximum of curvature

$$\overline{PI_{MC}} = \kappa_{max} \tag{27}$$

where κ is the trajectory curvature [34]. The lower the value of the performance index $PI$is, the better performance the driver-controller control system obtains.

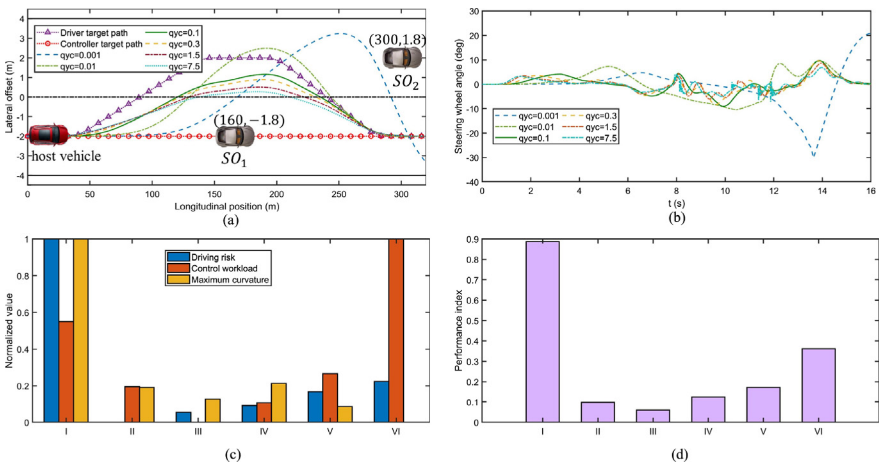

Fig. 5. Simulation results of determining the suitable lateral position error weight of the controller: (a) Illustration of the simulation scenario and vehicle trajectories, (b) Shared steering wheel angle, (c) Normalized value of three evaluated terms, and (d) Performance index. I: $q_{yc} = 0.001$, II: $q_{yc} = 0.01$, III: $q_{yc} = 0.1$, IV: $q_{yc} = 0.3$, V: $q_{yc} = 1.5$, and V: $q_{yc} = 7.5$.

4.3.2. Parameter selection

By referring to [11] and simulation experience, there are six groups of the lateral position error weights of the controller q_{yc}, namely, 0.001, 0.01, 0.1, 0.3, 1.5, and 7.5. Besides, six groups are named from I to VI, respectively. The driving scenario designed for selecting the weight of the controller is shown in Fig. 5(a). There are two static obstacle vehicles on the road, and the initial lateral position of the first obstacle vehicle SO_1 is at the middle on the right-hand lane and 160 m far from the host vehicle. Besides, the initial lateral position of SO_2 is at the middle on the left-hand lane and 300 m far from the host vehicle. The vehicle runs at the steady speed of 20 m/s. Assume that a moderate human driver has observed these two obstacles and executes the lane-changing maneuvers to avoid obstacles, however, the automated driving system occurs unexpected failure and tries to keep the vehicle going straight. The target paths of the trajectory-following controller and a human driver are illustrated in Fig. 5(a), and vehicle trajectories of six group weights are also presented in Fig. 5(a). Compared with other groups, the host vehicle controlled under the controller's weight of 0.001 performs badly since the trajectory curvature is large and there is under high collision risk. Six groups of shared steering wheel angle are shown in Fig. 5(a), which are the combination of the human driver's and the controller's steering control signal under the Nash equilibrium condition. As we can observe, there are the wide range and variation of the shared control signal under the controller's weight of 0.001. Therefore, the weight of 0.001 is eliminated. Besides, the shared control signal occurs sharp variations under the weight of 7.5, and the weight of 7.5 could also be eliminated. Fig. 5(c) shows the normalized values of three evaluated terms among six weight groups, and the performance index *PI*is presented in Fig. 5(d). As a result, when the lateral position error weight of the controller is set at 0.1, the shared system obtains the lowest value of the performance index *PI*. Therefore, the lateral position error weight of the controller is set at 0.1 in our proposed strategy.

5. Case studies

In order to validate the proposed dynamic authority allocation strategy, two case studies are provided in this section. The road shape and obstacle condition are considered in the design of tested scenarios, and listed detailedly as follows.

(1) There are three obstacles with different initial states on a straight road. After exceeding the last obstacle, there is a long segment of road without obstacles, which is designed as the safe driving condition. The main purpose of this scenario is to discuss the proposed dynamic authority allocation strategy under different driving risk conditions.
(2) There are two obstacles with different initial states on a curvy road. One obstacle is the static obstacle and the other is a dynamic obstacle. The objective of this scenario is to study the feasibility of the proposed strategy under different road shapes.

5.1. Experimental hardware/software

The driver-in-the-loop simulation platform is shown in Fig. 6. Logitech G27 is provided for inputting the control actions of the human driver, including the steering wheel angle, throttle opening, and braking pedal displacement. The traffic scenarios are designed by the *CarSim* software, which is a commercial vehicle dynamics software, and they could be visualized by the *CarSim-LabView* simulation environment in real-time. Besides, the E-class Sedan vehicle model provided by *CarSim* software is selected for the vehicle simulation and the design of the game-based shared control. The parameters of the vehicle model are also provided in Table 3.

Fig. 6. Driver-in-the-loop simulation platform.

Table 3
Parameters of Vehicle Model and Case Studies.

Symbol	Description	Value(Units)
m	Mass of the host vehicle	1650(kg)
l_a	Distance from CoG to the front axle	1.40(m)
l_b	Distance from CoG to the rear axle	1.650(m)
I_z	Vehicle yaw moment of inertia	3234(kg·m^2)
J_w	Wheel inertial	2.9(−)
G	Ratio of the steering system	18(−)
r	Wheel effective radius	0.36(m)
N_p/N_u	Prediction/ Control horizon length	100/100(−)
$q_{\varphi c}$	Yaw angle error weight of the controller	0(−)
$q_{\varphi d}$	Yaw angle error weight of the human driver	0(−)
q_{yc}	Lateral position error weight of the controller	0.1(−)

5.2. Participants and procedure

6 drivers, 4 males and 2 females, were recruited to participate in our study. Their average driving experience was 35.3 months with a standard deviation of 12.5 months. During the case studies, participants should follow the rules listed below.

(1) All participants were in the normal driving state. Drowsy or distracted driving states were not allowed, such as using smart phone while driving.
(2) The information of the proposed shared control strategy was illustrated to all participants.
(3) Each participant was allowed to finish four training tests of each scenario to be familiar with the driving-in-the-loop simulation platform.

All participants were required to conduct 11 experiments in total in each testing scenario under three driving conditions: (1) 5 experiments under the driver alone driving condition, (2) 3 experiments under the scaled constant intervention condition where the lateral position error weight of the human driver q_{yd} and the controller q_{yc} are both set to 0.1, and (3) 3 experiments with the proposed game-based shared control. In our study, the average path among the vehicle paths of 5 experiments under driver alone driving condition is chosen as the human driver's target path. As shown in Fig. 7, the vehicle paths under driving alone condition are consistent while driving at the same scenario, and both their average paths are close to the vehicle paths under driving alone condition. Therefore, the average path is used as the target path of a human driver under the normal driving state. Simultaneously, in order to present the human–machine conflicts during obstacle avoidance and depict the results of control authority allocation distinctly, the obstacle avoidance path of the trajectory-following controller is planned relative aggressive by the trajectory planning algorithm based on the accurate information of obstacles and roads when compared with the target path of a human driver. Besides, an example of each case study is provided to discuss among the driver alone, scaled intervention, and the proposed shared control in the following parts. Specifically, the example experiment results of three driving conditions are provided from the same driver in two testing scenario. Three evaluated items of the performance index are normalized among the experiment's data from a single driver for considering the drivers' characteristic differences.

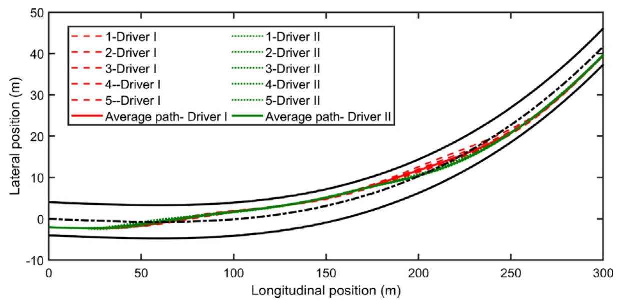

Fig. 7. An example of a human driver's target path.

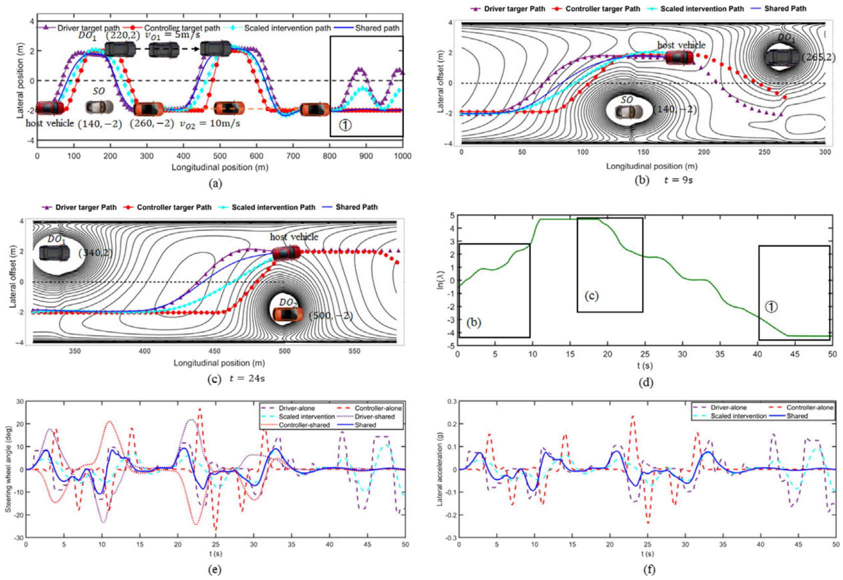

Fig. 8. Results of scenario I on a straight road: (a) Illustration of scenario I, the target path of a driver and the controller, and vehicle paths under dynamic authority allocation strategy and the scaled intervention, (b) Presentation of the driving safety field and vehicle paths at simulation time t = 9s, (c) Presentation of the driving safety filed and vehicle paths at simulation time t = 25s, (d) Authority allocation ratio, (e) Steering wheel angle, and (f) Lateral acceleration. Marker points are the preview path points, and solid lines are the actual paths in (b) and (c). *SO*means the static obstacle, DO_1 and DO_2 refer to the first and second dynamic obstacles.

5.3. Scenario I

The scenario I is illustrated in Fig. 8. There are two road segments in the scenario I and three obstacles are designed on different lanes, but there is no obstacle ahead on the road after the host vehicle moves over 800 m (simulation time t = 40s). The host vehicle runs with a steady speed as 20 m/s and its intimal position is at the middle on the right-hand lane. As shown in Fig. 8(a), the first obstacle *SO*is a static obstacle, which is 140 m far from the initial position of the host vehicle and at the middle point of the right-hand lane. The next obstacle is a dynamic obstacle vehicle DO_1, whose initial position is 220 m far from the host vehicle, and it moves along the middle on the left-hand lane at the steady speed of 5 m/s. Besides, the other dynamic obstacle vehicle DO_2 moves along the middle line of the right-hand lane at the steady speed of 10 m/s, and its initial position is 260 m far from the host vehicle.

Moreover, the target paths of a driver and the controller, and the vehicle paths of the scaled intervention and our proposed shared strategy are also presented in Fig. 8(a). As we could observe, the shared path is smooth and the host vehicle under the proposed strategy could avoid obstacles safely. Besides, as listed in Table 4, the most participants under the proposed shared control get the lowest average normalized evaluated values of the maximum curvature when compared with the condition of driver alone and the scaled constant intervention. Fig. 8(b) shows a detailed path result from simulation time t = 0s to t = 9s, and Fig. 8(c) shows the result from t = 16s to t = 24s. When the position of the host vehicle is between two dynamic obstacles, two dynamic obstacles both have the influence on the driving risk of the host vehicle in Fig. 8(c). However, the dynamic obstacle DO_1 has little influence on the host vehicle because of the large relative distance in Fig. 8(b) when compared with the condition in Fig. 8(c). It is more dangerous to avoid obstacles under the condition in Fig. 8(c) than the latter. Therefore, the human driver should occupy a higher control authority and the authority allocation ratio λ should be larger under the condition in Fig. 8(c), as shown in the rectangle marked with (b) and (c) in Fig. 8(d). And the solid shared path in Fig. 8(c) is closer to the human driver path than in Fig. 8(b). As depicted in shown in Fig. 8(e), the driver's and controller's steering curve under shared control driving condition grow in opposite directions because of the interaction between a driver and the trajectory-following controller during the dynamic game. Two control participants of the host vehicle, a driver and the controller, adopt their control strategies by considering the other's steering action, as illustrated in Fig. 1.

After the host vehicle exceeds the dynamic obstacle DO_2, there is no obstacle ahead and it is a safe driving condition, which is marked by rectangle ① in Fig. 8(a). According to the proposed dynamic authority allocation strategy, the host vehicle could be controlled by the trajectory-following controller under the guidance of the controller's target path to decrease the driver's workload. In order to show the control authority clearly in ① marked rectangle area, a participant is required to control the vehicle by random control actions after t = 40s. From the ① marked rectangle area in Fig. 8(d), the nature logarithm value of authority allocation ratio ln(λ) gets a negative large value and it means the controller obtains full control authority. Therefore, the shared path is very close to the controller's target path shown in the ① marked rectangle in Fig. 8(a) and (e). Therefore, as presented in Table 4, all participants under the proposed shared control condition obtain

Table 4
Average Normalized Values of Three Evaluated Terms and Performance Index in Scenario I.

	Driver I				Driver II			
	DR	CW	MC	PI	DR	CW	MC	PI
DA[1]	**0.20**	0.86	0.63	0.47	**0.28**	0.89	0.80	0.56
SI[2]	0.60	0.39	0.11	0.42	0.83	0.25	0.11	0.51
SC[3]	0.61	**0.05**	**0.01**	**0.32**	0.43	**0.05**	**0.06**	**0.24**
	Driver III				Driver IV			
	DR	CW	MC	PI	DR	CW	MC	PI
DA	**0.31**	0.82	0.85	0.57	**0.25**	0.69	0.53	0.43
SI	0.88	0.19	0.17	0.53	0.92	0.20	**0.05**	0.52
SC	0.77	**0.00**	**0.06**	**0.40**	0.76	**0.03**	0.07	**0.40**
	Driver V				Driver VI			
	DR	CW	MC	PI	DR	CW	MC	PI
DA	**0.25**	0.78	0.67	0.49	**0.23**	0.79	0.66	0.48
SI	0.83	0.20	0.04	0.47	0.89	0.32	0.21	0.57
SC	0.62	**0.04**	**0.01**	**0.33**	0.71	**0.01**	**0.04**	**0.37**

[1] Driving alone.
[2] Scaled intervention.
[3] Shared control.

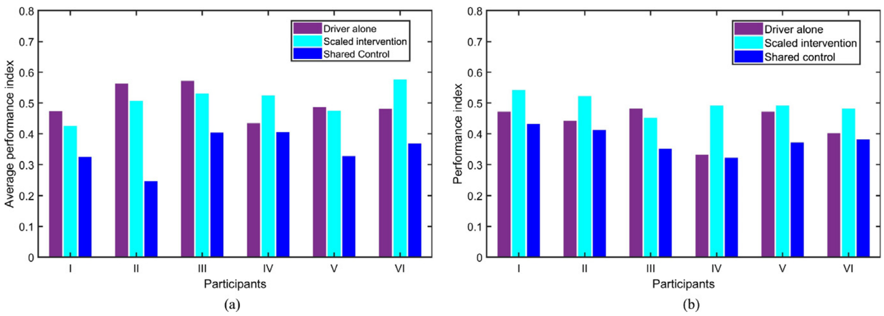

Fig. 9. Average normalized values of three evaluated terms and the performance index of each participants under three driving conditions: (a) scenario I and (b) scenario II.

the lowest average normalized values of the control workload, and the normalized values of the control workload are significantly decreased when compared with the other two driving condition. Moreover, as presented in Fig. 8(f), the host vehicle with the proposed strategy obtains great vehicle stability performance since the absolute maximum value of lateral acceleration is lower than 0.1 g during obstacles avoidance.

Importantly, Fig. 9(a) evaluates the performance index for six participants under three driving conditions in scenario I. It reveals that the proposed shared control algorithm obtains the lowest *PI*value for all participants, which means the proposed shared control algorithm gets the better performance than the driver alone and the scaled constant intervention condition in a straight scenario.

5.4. Scenario II

As shown in Fig. 10(a), there are two obstacles on a curvy road and the parameters of a curvy road are also presented at the top and left rectangle. The initial position of the static obstacle *SO*is at the middle on the right-hand lane and 120 m far from the host vehicle. Besides, there is a dynamic obstacle running alone the middle on the left-hand lane at a steady speed as 10 m/s on a curvy road, whose initial longitudinal position is 150 m far from the host vehicle. The host vehicle moves at a steady speed as 20 m/s.

The target paths of a human driver and the controller, and the vehicle paths under the scaled constant intervention and the proposed shared strategy are also depicted in Fig. 10(a). Besides, the detailed path results from t = 0s to t = 4s and from t = 6s to t = 12s are shown in Fig. 10(b) and (c), respectively. In this case study, the shared path is smooth and close to the target path of the human driver. And the vehicle under the proposed dynamic authority allocation strategy could avoid obstacles safely. Moreover, the elliptic driving safety filed is also able to adapt to the curvy road from Fig. 10(b) and (c). From Fig. 10(d), the authority allocation ratio λ of Fig. 10(b) and (c) marked with rectangles are similar with the conditions of Fig. 8 (b) and (c), and the driving risk is larger at the condition that the position of the host vehicle is between two obstacles in (c)

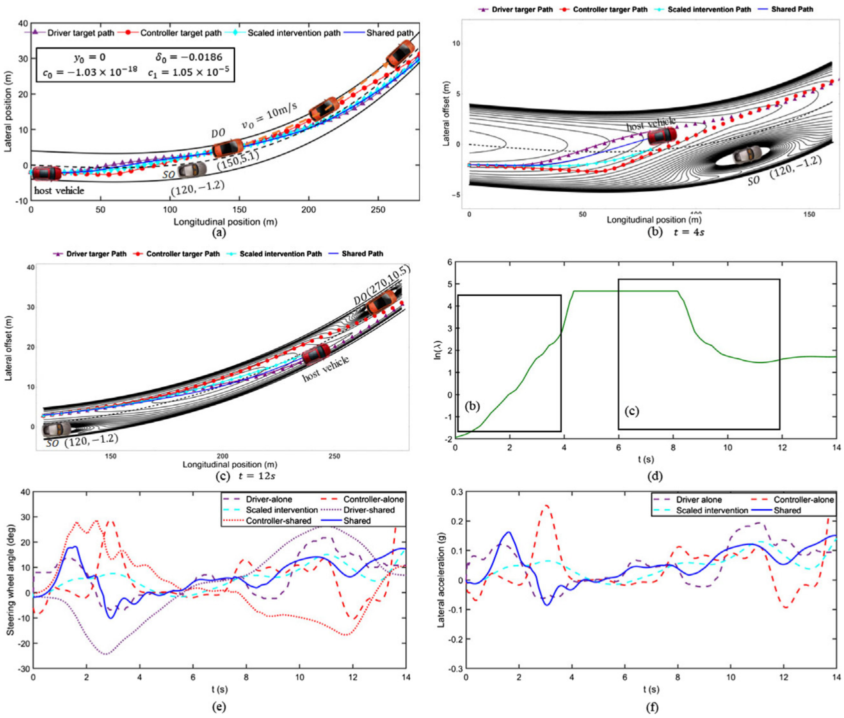

Fig. 10. Results of scenario II on a curvy road: (a) Illustration of scenario II, the target path of a driver and the controller, and vehicle paths under dynamic authority allocation strategy and the scaled intervention, (b) Presentation of the driving safety filed and vehicle paths at simulation time t = 4s, (c) Presentation of the driving safety filed and vehicle paths at simulation time t = 12s, (d) Authority allocation ratio, (e) Steering wheel angle, and (f) Lateral acceleration. Marker points are the preview path points, and solid lines are the actual paths in (b) and (c).

Table 5
Average Normalized Values of Three Evaluated Terms and Performance Index in Scenario II.

	Driver I				Driver II			
	DR	CW	MC	PI	DR	CW	MC	PI
DA	**0.31**	0.65	0.61	0.47	**0.13**	0.74	0.74	0.44
SI	0.88	**0.14**	0.27	0.54	0.93	**0.07**	**0.15**	0.52
SC	0.55	0.45	**0.15**	**0.43**	0.45	0.51	0.23	**0.41**
	Driver III				Driver IV			
	DR	CW	MC	PI	DR	CW	MC	PI
DA	**0.23**	0.62	0.83	0.48	**0.11**	0.50	0.59	0.33
SI	0.85	**0.03**	**0.08**	0.45	0.90	**0.04**	**0.11**	0.49
SC	0.35	0.39	0.32	**0.35**	0.42	0.21	0.21	**0.32**
	Driver V				Driver VI			
	DR	CW	MC	PI	DR	CW	MC	PI
DA	**0.09**	0.88	0.83	0.47	**0.15**	0.79	0.52	0.40
SI	0.89	**0.08**	**0.11**	0.49	0.90	**0.09**	**0.03**	0.48
SC	0.46	0.28	0.28	**0.37**	0.36	0.56	0.22	**0.38**

marked area than the condition in (b) marked area, therefore, the value of $\ln(\lambda)$ in the former area is larger than the latter. The shared path in Fig. 10(c) is closer to the target path of the human driver than the shared path in Fig. 10(b). As depicted in Fig. 10(b) and Table 5, the vehicle under the scaled intervention driving condition obtains the lowest control workload with the cost of the maximal driving risk. Besides, the vehicle with the proposed strategy also could obtain great vehicle stability performance on a curvy road as the absolute maximum value of lateral acceleration is 0.16 g that is lower than 0.4 g, which is shown in Fig. 10(f). Moreover, Fig. 9(b) shows the average performance index of six participants under three driving conditions in the scenario II. According to Fig. 9(b) and Table 5, the proposed shared control could obtain better performance than the scaled constant intervention and driver alone condition since the performance indexes of six participants get the lowest values when compared to other two driving conditions.

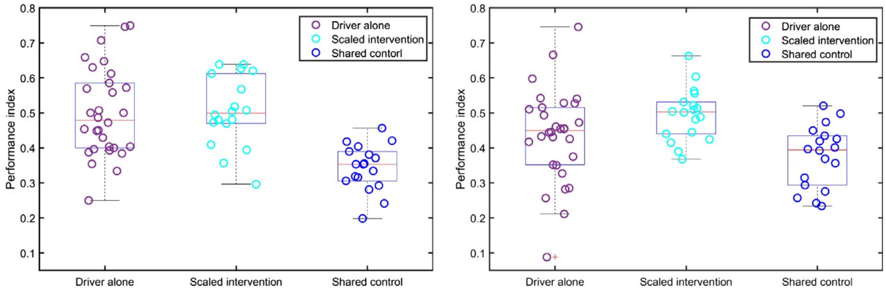

Fig. 11. Boxplot of performance indexes under three driving conditions: (a) scenario I and (b) scenario II. Circular points represent the original data.

Importantly, the boxplots of the six participants' performance indexes under three driving conditions in the scenario I and II are shown in Fig. 11. It proves that the proposed shared control could help the driver obtain better performance in different road shapes and driving conditions.

6. Discussion and conclusion

This paper proposes a novel dynamic control authority allocation strategy for the game-based shared control to deal with human-automation conflicts. The dynamic authority allocation strategy between a driver and the trajectory-following controller is designed based on the driving risk evaluated by the elliptic driving safety field, and the driving safety filed is constructed under the consideration of the driver-vehicle-road interactions. The human-automation interaction is modeled by the noncooperative game theory, where the Nash equilibrium solution is derived by the DMPC approach. The dynamical authority allocation could be realized by tuning the cost functions of a human driver and the trajectory-following controller according to the authority allocation ratio. Case studies results demonstrate that the control authority is gradually transferred to a human driver under the high driving risk condition, on the contrary, it is fully transferred to the trajectory-following controller to relieve the driver's control workload. Moreover, the vehicle driven by the proposed strategy could avoid obstacles safely and obtains great stability performance under different driving scenarios.

The dynamic control authority allocation strategy is studied on the assumption of accurate data information. Especially, the driving safety field is constructed based on the information of the environment (e.g. obstacles, road lanes), vehicle (e.g. position, velocity), and driver characteristic (e.g. driving skill, driving style). Therefore, the accuracy and access to the above information have great influence on the evaluated driving risk and the proposed strategy. However, the development of advanced sensor fusion technology [24,25,35] and V2X transportation systems [36] provide possibilities to obtain the accurate environment and vehicle state information. For example, obstacle detection could be well addressed by the Extended Kalman Filter (EKF) method based on multi-sensors of radar, Lidar, and camera [24]. Besides, the driver characteristics could also be recognized by machine learning methods with high accuracies, such as driving style [37], driving skill level [38], and driving distraction [39]. Therefore, it's certainly possible to obtain higher accuracy information for the proposed strategy, and our proposed method is feasible and has a higher potential for real application. Moreover, the average path among paths under driver alone condition is selected as the human driver's target path. When the proposed shared control is applied into commercial, the target path of a human driver could be obtained by adaptive learning method from driver's daily driving data [40].

In future work, the game-based shared control will be implemented into an advanced driving simulator for taking a real driver in the control loop and considering the response of a human driver to the proposed dynamic control authority allocation strategy. Besides, the driver's trust in automation and response to the proposed share control systems are the potential influence factors of a driver's target path with the consideration of human–machine interaction [41,42]. Therefore, a suitable method to predict the driver's target path in real time should be studied by considering driver's trust in automation and response. Furthermore, in order to obtain safer and more effective performance, the driver status, such as drowsy or distracted driving, driver's control authority intention, and the suitable warning mechanism should be added in the design of the proposed strategy.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, athttps://doi.org/10.1016/j.ymssp.2019.01.040.

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